

# Virtual Manipulation of Cultural Assets: An Initial Case Study with Single-Joint Articulated Models

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## ABSTRACT

Virtual space can eliminate the physical constraints of real space. Three-dimensional (3D) digital twins enable us to experience manipulation of cultural assets that are inaccessible in real space. However, most 3D models obtained using current reconstruction techniques are static; they cannot be manipulated dynamically. In this study, we reproduce a dynamic 3D model of a simple articulated object with a rotating joint from a static point cloud only with reference to a video of the motion and a manually added rotation axis. The reconstructed actual cultural asset is used for first-person experiences in an augmented reality environment.

## CCS CONCEPTS

• Computing methodologies → Shape analysis.

## KEYWORDS

point clouds, segmentation, 3D reconstruction, rotational motion, articulated objects

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## 1 INTRODUCTION

Recently, various content has been created for the metaverse, which can provide unprecedented experiences without the physical constraints of real space. In particular, cultural assets are generally prohibited from direct interaction for protection; hence, interactive exhibitions that enable us to experience manipulation of cultural assets using three-dimensional (3D) digital twins can offer significant benefits. However, most 3D models obtained using current reconstruction techniques are static; they cannot be manipulated dynamically. Continuous point cloud capture does not guarantee accurate content recognition in dynamic scenes due to the lack of point correlation maintenance between frames. In addition, an RGBD camera is necessary. Practical virtual exhibitions should be feasible using readily available photography equipment.

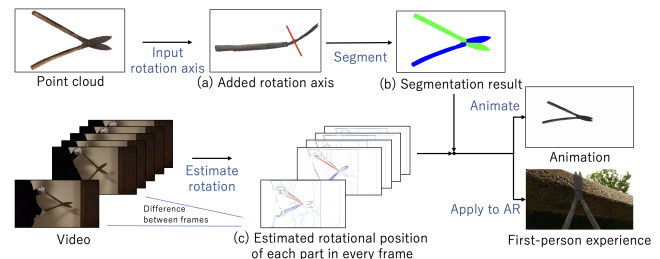
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**Figure 1: Processing flow of the proposed method. The target object is a simple articulated object consisting of two parts joined by a 1DoF rotational joint.**

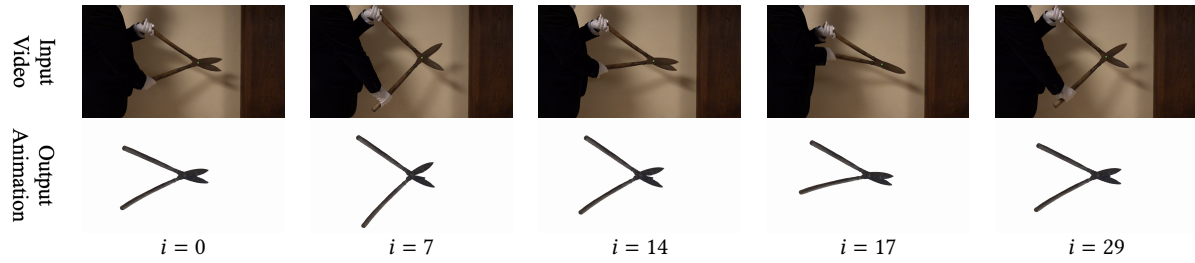
A more effective approach to represent a 3D reconstructed dynamic scene is to understand the structure and motion of the object and adaptively deform the point cloud. The method in [Yi et al. 2018] learns the interpoint correspondence for each point cloud in a different state during joint motion to segment the point cloud into two parts. However, because this approach requires two point cloud datasets, the cost of the reconstruction is considerable. The method in [Li et al. 2020] detects the shape and articulation of parts using deep learning with labeled point cloud datasets. However, due to its reliance on training data for each category, this method is not suitable for data-poor cultural assets.

In this study, we reproduce a dynamic 3D model of a simple articulated object with a rotating joint from a static point cloud. For starters, we reconstruct such a model only with reference to a video of the motion and a manually added rotation axis. The proposed method involves part segmentation and estimates the rotational position at each frame from the input video to reproduce the joint motion. This method does not rely on an RGBD camera, and the generated models can be easily utilized as the dynamic content for an augmented reality (AR) or virtual reality environment.

## 2 METHOD

Figure 1 shows the processing flow of the proposed method. First, the input point cloud is segmented into two parts with reference to the specified rotation axis. Next, the rotational position of each part in each video frame is estimated. Finally, the joint motion is reproduced with these segmented parts and the estimated rotational positions.

The rotation axis specification, shown in Fig. 1(a), allows the user to freely move the viewpoint and set the starting and ending points of the rotation axis by clicking on each point in the point cloud. The 3D coordinates of each end of the rotation axis are obtained from the projected coordinates of the clicked point.



**Figure 2: 3D model animation of an actual cultural asset, where  $i$  denotes the frame number of the input video.**

The point cloud is then segmented into two parts, as shown in Fig. 1(b), based on the specified rotation axis. Two planes, including the rotation axis, are detected using random sample consensus (RANSAC) [Schnabel et al. 2007], which is a commonly used shape detection method for point clouds. Then, the neighboring points of the two planes are classified into either of the planes. After that, the neighboring points of the classified points are classified into the same plane. This process is repeated until the object has no unclassified points.

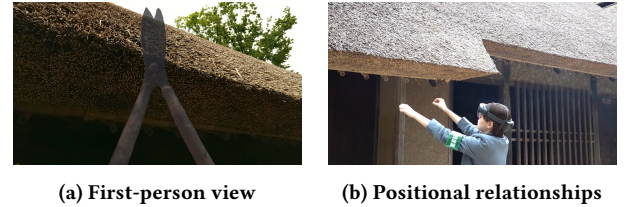
Next, the rotational position of each part is estimated from the video, as shown in Fig. 1(c). The optical flow of rotational motion forms a perfect circle. We measure the angle between the line segment from the center to the starting point and the line segment from the center to the ending point of the optical flow. Subsequently, optical flows are clustered by hierarchical clustering analysis to distinguish the movement of each part in terms of two variables: (1) angular declination  $\theta$  of the starting point relative to the rotational axis and (2) the rotation angle  $d\theta$  between the starting and ending point. The number of optical flows in the cluster and the difference between the angular declination of the starting point and that of the previous frame are used to identify the two representative clusters. The average of  $d\theta$  in the selected cluster is used as the amount of angular change of each part in the frame and is added to the previous rotational position.

Finally, animations with an unfixed viewpoint are generated by rotating each part independently around the rotation axis with the estimated angle. The Poisson surface reconstruction [Kazhdan et al. 2006] is utilized to impose a mesh on the point cloud, resulting in a continuous appearance.

### 3 RESULTS

Figure 2 excerpts five frames from an animation produced by executing the proposed method on the actual cultural asset of an old folk tool. We visited the Japan Open-air Folk House Museum (<https://www.nihonminkaen.jp>) to take multi-view images of roof shears. A point cloud was reconstructed from these images using LumaAI (<https://lumalabs.ai>). Roof shears are used to trim the thatch on thatched roofs. Preprocessing eliminated optical flows other than the joint motion by tracking a marker placed on the rotation axis. Using the synthesized dynamic model, a simulated experience of manipulating the cultural asset was presented by adding a hand-tracking operation interface on Microsoft HoloLens 2 (<https://www.microsoft.com/hololens>). The range of rotational motion was constrained from the minimum to maximum rotational position in the input video. The simulated experience is shown in

Fig. 3, where (a) shows a first-person view while (b) represents the positional relationships among the user, the virtual shears, and the roof. The result shows that the interactive 3D model can represent the dynamic functionality of the tool. See the supplementary video for more details.



**Figure 3: First-person experience in an AR environment using the resulting interactive 3D model.**

### 4 CONCLUDING NOTES

In this study, we proposed a method to reconstruct joint motion from a single point cloud, together with an RGB video and a user-specified rotation axis. In addition, we demonstrated that the resulting 3D model can be easily applied to first-person simulated experiences in an AR environment.

In future work, we should consider a method to estimate the joint motion of more complex objects with multiple rotation axes. We believe that utilizing the video for the part segmentation process will allow us to handle a greater variety of articulated objects. For the input video, we should eliminate the restriction of positioning a camera on the rotation axis. Investigating deep learning-based reconstruction approaches is also an issue of interest.

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